

ESE 101:

Homework 3 (due October 30, 2025):

1 Greenhouse Effect

Consider an atmosphere consisting of n homogeneous layers in radiative equilibrium with an underlying surface. Assume the atmosphere absorbs no solar radiation, the surface has albedo α , and the atmospheric layers absorb and emit as blackbodies (emissivities $\epsilon = 1$).

1. What is the radiative energy balance at the top of the atmosphere and in each atmospheric layer?
2. What is the radiative equilibrium temperature for each layer and the surface as a function of n ? What are representative values for Earth in the top layer of the atmosphere and at the surface for a few values of n (total solar irradiance $S_0 = 1362 \text{ W m}^{-2}$ and albedo $\alpha = 0.3$)?
3. Provide a (narrative) physical explanation for the dependence (or lack thereof) of the temperature of the top layer and of the surface on n .
4. The atmospheric layer adjacent to the surface may be physically thin. Yet there is a difference between the temperature of the surface and the temperature of this near-surface layer of the atmosphere. What causes it? What are some of its implications?

2 Pressure Broadening

One of the ways radiation can interact with matter is by ABSORPTION. For a given adsorption line, the wavelengths/frequencies of light that can cause a molecule to undergo specific energetic transitions (e.g. electronic or vibrational) follow a distribution (in this context it is usually called a SPECTRAL LINE SHAPE). The width of one of these line shapes is determined by the finite lifetime of the excited state of a molecule. In this question, we will be thinking about how

changes in pressure change these line shapes. From class, recall the Lorentz Line Profile:

$$\Phi_L(\nu, \alpha) = \frac{1}{\pi} \left[\frac{\alpha}{(\nu_0 - \nu)^2 + \alpha^2} \right]$$

Here, ν_0 is the peak frequency for the line shape and α is the HALF-WIDTH (the distance away from ν_0 when half of the peak height is achieved). We use the Lorentz line shape because it is the appropriate shape to describe spontaneous decay. We will save you from all of the math (see Canvas for PDF), but using the ideal gas law, Boltzmann's distribution, and Plank's Law, we can write the following expression for line half-width:

$$\alpha = \alpha_0 \left(\frac{p}{p_0} \right) \sqrt{\frac{T_0}{T}}$$

If we ASSUME the troposphere is adiabatic (this assumption gives us a way to link temperature and pressure in the atmosphere), we can further assume $T_0/T = (p/p_0)^{\frac{1-\gamma}{2\gamma}}$ (where $\gamma = c_p/c_v$ is the heat capacity ratio) and make a further substitution and get:

$$\alpha = \alpha_0 \left(\frac{p}{p_0} \right)^{\frac{1+\gamma}{2\gamma}} \quad \text{or} \quad \alpha = \alpha_0 \left(\frac{T}{T_0} \right)^{\left[\frac{\gamma+1}{2(\gamma-1)} \right]}$$

This equations tells us that if we know/measure (T_0, p_0, α_0) then I can calculate a new α for a given (T, p) .

1. Calculate the ratio α/α_0 of half-widths for water at the surface (high pressure) $T_0 = 20^\circ\text{C}$ and water higher in the atmosphere (low pressure) $T = 1^\circ\text{C}$ (take $\gamma = 1.33$) and plot the Lorentz line shape for both temperatures as a function of multiples of α_0 . What does this tell you about the line widths for water near the surface v. high in the troposphere?
2. Explain two reasons why the UPPER atmosphere is generally transparent to radiation.

3 More Radiation

Suppose today is September 22, 2023 (the Fall equinox) and you are in Pasadena, CA (34.15°N , 118.14°W).

1. Using code from last week and the textbook to guide you, calculate and plot the solar zenith angle as a function of the time of day. You may assume solar noon occurs at 12pm (noon). Label all axes and put a title on the plot (**for all plots in the HW set**).
2. Now, make a plot of the top of atmosphere (TOA) insolation at this location as a function of time of data.
3. In the accompanying data file, you will find observed temperature and insolation recorded at Caltech (on the roof of Linde Laboratory to be specific). Make a plot of the observed insolation on the same plot as the TOA insolation you modeled above. What differences do you notice? Explain some possible reasons for this discrepancy. You may assume solar noon is the same as local noon since these weather reports use local time. In reality, solar noon was at 12:45pm local time on this day. You may account for this in your plots if you so choose.
4. Now, use the observed temperatures in the accompanying data file and the observed insolation on the Linde roof to make a plot of the estimated outgoing longwave radiation and compare it to the incoming shortwave radiation. Explain what you see in 2–3 sentences.
5. You might notice that the outgoing longwave radiation integrated over the day appears to be much more than the incoming shortwave radiation (feel free to take an integral to verify this!). But the temperatures appear relatively stable (i.e., temperatures aren't rapidly falling in the afternoon). Why might this be the case?